

FROGS AND TOADS

With powerful hind limbs tucked under its body, a wide mouth, and protruding eyes, the typical frog has a unique and unmistakable body shape.

Anura, the name of the order of frogs and toads, means “animals without a tail”. As adults, all other amphibians have a tail, but in the anurans, the tail gradually disappears during metamorphosis from the larval to the adult stage of the life cycle. The larvae, known as tadpoles, feed chiefly on plant material and have a spherical body that contains the long, coiled gut required by such a diet. The adults, in contrast, are totally carnivorous, feeding on a wide range of insects and other invertebrates, with larger species also taking small reptiles and mammals, as well as other frogs.

INGENIOUS ADAPTATIONS

Frogs and toads catch their prey by ambush, most spectacularly by jumping. In many frogs and toads, the hind limbs are modified for jumping, being much longer than the forelimbs and very muscular. Jumping is also an effective method of escaping from predators, of which frogs and toads have many.

Not all anurans jump, however. Many have hind limbs adapted for other kinds of locomotion, including swimming, burrowing, climbing, and, in a few species, gliding through the air. Most frogs and toads live in damp habitats, close to the pools and streams in which they breed, but there are several species adapted to life in very arid areas. The greatest diversity of anurans is found in the tropics, particularly in rainforests. Many are active by day, while others are nocturnal. Some species are cleverly camouflaged; in contrast, others are brightly coloured, advertising that they are poisonous or unpleasant to taste.

COURTSHIP AND BREEDING

Anurans differ from other amphibians in having a voice and very good hearing. Males of most species call to attract females, making sounds characteristic of their species. In all but a very few species, fertilization is external, the male shedding sperm onto the eggs as they emerge from the female's body. To do this, the male clasps the female from above – a position known as amplexus. The duration of amplexus varies between species – from a few minutes to several days.

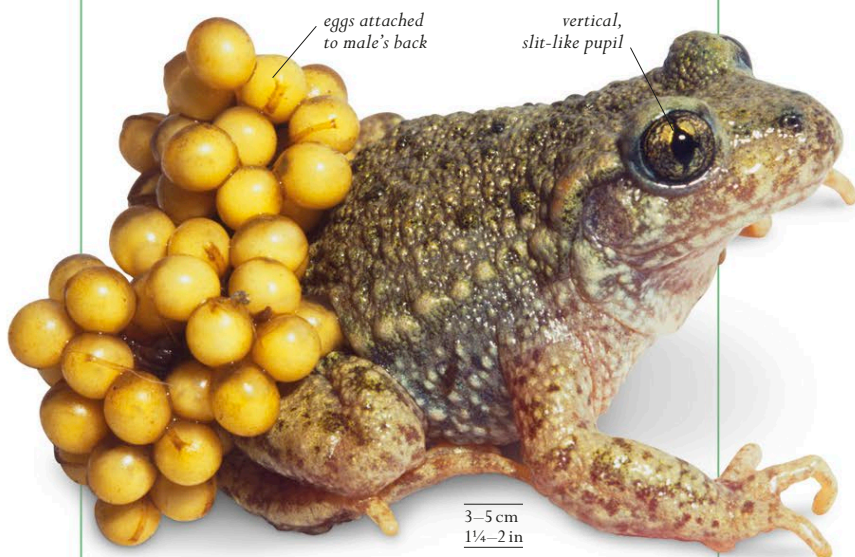
PHYLUM	CHORDATA
CLASS	AMPHIBIA
ORDER	ANURA
FAMILIES	54
SPECIES	7,244

DEBATE FROG OR TOAD

The distinction between frogs and “toads” is biologically meaningless and the two words are used differently in different parts of the world. In Europe and North America, for example, the term “toad” refers to species in the family Bufonidae, but this also includes the South and Central American harlequin frogs. In general, toads have rough skins, are slow-moving, and often burrow in the ground, whereas frogs are smooth-skinned, agile, and fast-moving, and spend much of their time in water. A group of smooth-skinned, aquatic frogs native to Africa were once known as clawed toads, but are now called clawed frogs.

ALYTIDAE

Males of the Alytidae, a small family of terrestrial frogs, call at night to attract females. When mating, the male midwife toad attaches fertilized eggs to his back. He carries them until they are ready to hatch, when he releases the tadpoles into water. Occasionally, a male may carry egg strings from more than one female. With painted frogs, a female mates with several males, producing up to 1,000 eggs, which are dropped into water.



COMMON MIDWIFE TOAD
Alytes obstetricans

Found in W. and C. Europe, this midwife toad has a plump body and powerful forelimbs adapted for digging. It hides in a burrow by day.

SQUEAKERS AND RELATIVES

The Arthroleptidae are a large, diverse family of frogs occurring across sub-Saharan Africa – in forest, woodland, and grassland, some at high altitudes. Its members range from tiny “squeakers”, named for their high-pitched call, to large treefrogs.



**RIO BENITO
LONG-FINGERED
FROG**

Cardioglossa gracilis
An inhabitant of lowland forest, this frog breeds in streams. Males call from nearby slopes.



**WEST CAMEROON
FOREST TREEFROG**
Leptopelis nordequatorialis

This large treefrog lives in montane grasslands of W. Africa. Males call to females near water when breeding, and eggs are deposited in ponds or marshes.



**WEST AFRICAN
SCREECHING FROG**

Arthroleptis poecilonotus
The female of this small species lays large eggs in soil cavities. The male is noted for his loud call.



AFRICAN TREEFROG
Leptopelis modestus

This species is found near streams in forests of W. and C. Africa. Females are larger than males.